

for $(u, v) \in \ker d_x F(x, y) \times \ker d_y F(x, y)$ at every $(x, y) \in \partial\Sigma \cap (U \times V)$. The argument is quite simple. By fixing boundary points (x, y) and vectors (u, v) as specified above, let $\gamma : [0, 1] \rightarrow \partial\Sigma^y \cap U$ be a curve with $\gamma(0) = x$ and $\gamma'(0) = u$, and set $h(s) = d_y F(\gamma(s), y)$. The curvature condition (2) means that $h(s) = \alpha(s)h(0)$ for some nonvanishing function $\alpha : [0, 1] \rightarrow \mathbf{R}$. In particular, we get

$$\langle d_x d_y F(x, y), u \otimes v \rangle = \langle h'(0), v \rangle = \alpha'(0) \langle h(0), v \rangle = 0.$$

Reciprocally, assume that the $\mathcal{C}^2$ -curvature condition above holds. Consider a curve $\gamma : [0, 1] \rightarrow \partial\Sigma^y \cap U$ and define h as above. Since we have $\gamma'(s) \in \ker d_x F(\gamma(s), y)$ and $h'(s) = \gamma'(s)^t \cdot d_x d_y F(\gamma(s), y)$ by construction, it turns out that $\langle h'(s), v \rangle$ equals $\langle d_x d_y F(\gamma(s), y), \gamma'(s) \otimes v \rangle$ for any $v \in \ker d_y F(\gamma(s), y)$. Applying the $\mathcal{C}^2$ -curvature condition, this implies that $h'(s)$ is parallel to $h(s)$ for every s , which leads to the ODE

$$\left. \begin{aligned} h'(s) &= \lambda(s)h(s) \\ h(0) &= d_y F(x, y) \end{aligned} \right\} \Rightarrow h(s) = \exp\left(\int_0^s \lambda(t)dt\right)h(0) = \alpha(s)h(0)$$

for a nonvanishing $\alpha : [0, 1] \rightarrow \mathbf{R}$. This implies condition (2) in Theorem A.

Remark 1.10. In this form, (2) is invariant under exchanging x and y , which is clear a posteriori without the $\mathcal{C}^2$ assumption, since both (1) and (3) are. On the other hand, condition (2) in Theorem A seems new, while its $\mathcal{C}^2$ -form above is quite similar to the *rotational curvature* $\det[d_x d_y F(x, y)]$ defined by Stein in [43, XI.3.1].

1.7. On the transversality condition. The transversality assumption has been essential in our proofs of (1) $\Rightarrow$ (2) $\Rightarrow$ (3) in Theorem A, but it is not clear that it is really needed for the statement. Indeed, conditions (1) and (3) make sense without it, and (2) is already meaningful if one only assumes that $\mathbf{n}_2(x_0, y_0) \neq 0$, and we do not have an example where the equivalence fails. It is likely that such examples can be found, but probably not for domains with analytic boundary. We leave these questions as open problems. In the degenerate case where $\mathbf{n}_1$ is identically 0, or equivalently when Σ is locally of the form $\Sigma = \{(x, y) : y \in \Omega\}$, all conditions in Theorem A hold. The S_p -boundedness is in that case even true for $1 \leq p \leq \infty$ because the Schur multiplier whose symbol is the indicator function of Σ is just the right-multiplication by the orthogonal projection on $L_2(\Omega)$.

2. Idempotent Fourier multipliers on Lie groups

Let G be a Lie group, that we equip with a left Haar measure. As to every locally compact group, we can associate to it:

- Its von Neumann algebra $\mathcal{L}G$.
- The noncommutative L_p spaces $L_p(\mathcal{L}G)$ for $1 \leq p < \infty$.
- The Fourier L_p -multipliers T_m with symbol $m : G \rightarrow \mathbf{C}$.

The group von Neumann algebra $\mathcal{L}G$ is the weak-* closure in $B(L_2(G))$ of the algebra of convolution operators $\lambda(f) : \xi \in L_2(G) \mapsto f * \xi$ for $f \in \mathcal{C}_c(G)$. When G is unimodular, its L_p -theory is quite elementary: $\mathcal{L}G$ carries a natural semifinite trace τ given by $\tau(\lambda(f)^* \lambda(f)) = \int |f(g)|^2 dg$ for every $f \in L_2(G)$ with $\lambda(f) \in \mathcal{L}G$; $L_p(\mathcal{L}G)$ is then defined as the completion of $\{x \in \mathcal{L}G : \|x\|_p < \infty\}$ for the norm $\|x\|_p = \tau(|x|^p)^{1/p}$. It turns out that $L_p(\mathcal{L}G)$ contains $\{\lambda(f) : f \in \mathcal{C}_c(G) * \mathcal{C}_c(G)\}$ as a